2.2.11

AI25BTECH11019 - MENAVATH SAI SANJANA

Question:
The plane $2x - 3y + 6z - 11 = 0$ makes an angle $\sin^{-1}(\alpha)$ with the x-axis. The value of α is equal to

Solution:

$$\vec{n} = \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix}, \quad \text{(normal vector of the plane)} \tag{0.1}$$

$$\vec{a} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \text{(direction vector of x-axis)} \tag{0.2}$$

$$\cos\phi = \frac{\vec{n} \cdot \vec{a}}{\|\mathbf{n}\|\|\mathbf{a}\|} = \frac{2}{\sqrt{2^2 + (-3)^2 + 6^2}} = \frac{2}{7} \tag{0.3}$$

$$\theta = 90^\circ - \phi \tag{0.4}$$

$$\sin\theta = \cos\phi = \frac{2}{7} \tag{0.5}$$

So, $\theta = \sin^{-1}\left(\frac{2}{7}\right)$

The plane $2x - 3y + 6z - 11 = 0$ makes an angle $\sin^{-1}(\frac{2}{7})$ with the x-axis. Hence, the required value of α is $\alpha = \frac{2}{7}$.

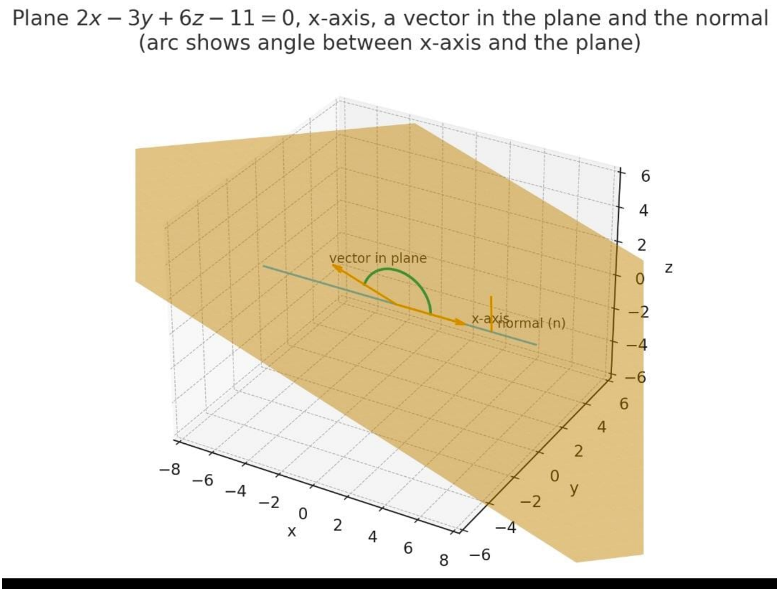
Plane $2x - 3y + 6z - 11 = 0$, x-axis, a vector in the plane and the normal
(arc shows angle between x-axis and the plane)
vector in plane
x-axis
normal (n)
x
y
z
−8 −6 −4 −2 0 2 4 6 8
−6 −4 −2 0 2 4 6
−6 −4 −2 0 2 4 6

Fig. 0.1